

Lecture 2

Comparing Loss Functions and Simple Linear Regression

DSC 40A, Summer 2026

Lecture 2 part 1

Comparing Loss Functions

Announcements

- Homework 1 is due to Gradescope on **Monday, July 6th**.
- Discussion is today, directly after lecture at 2pm here.
 - Remember that, in general, groupwork worksheets are released on Wednesday and due Thursday.
 - Starting next week you will need to sign up for a check off meeting with your group.
- Look at the office hours schedule [here](#) and plan to start regularly attending!
- Remember to take a look at the supplementary readings linked on the course website.
- Please participate in lecture through hand raising, chat, and/or Q&A.

Agenda for part 1

- Recap: Empirical risk minimization.
- Choosing a loss function.
 - The role of outliers.
- Center and spread.
- Towards linear regression.
- Break!

Follow-Ups and Questions from Lecture 1

- Are there guidelines to bad loss functions?
 - Kinda: if it fails to provide useful information or guides you towards predictions that don't align with the real-world goals.
 - Non-differentiable $\rightarrow$ we will explore why in more depth later on, but for now if the loss function isn't differentiable then there's no way to tell whether that point is a minimum.
 - Unbounded or exploding slope $\rightarrow$ we will be looking into this today!
- Any other lingering questions from last time?

Question 🤔

Answer in chat or Q&A

Remember, you can always ask questions.

Recap: Empirical risk minimization

Goal

We had one goal in Lecture 1: given a dataset of values from the past, **find the best constant prediction** to make.

$$y_1 = 72 \quad y_2 = 90 \quad y_3 = 61 \quad y_4 = 85 \quad y_5 = 92$$

Key idea: Different definitions of "best" give us different "best predictions."

The modeling recipe

In Lecture 1, we made two full passes through our "modeling recipe."

1. Choose a model.
2. Choose a loss function.
3. Minimize average loss to find optimal model parameters.

Empirical risk minimization

- The formal name for the process of minimizing average loss is **empirical risk minimization**.
- Another name for "average loss" is **empirical risk**.
- When we use the squared loss function, $L_{\text{sq}}(y_i, h) = (y_i - h)^2$, the corresponding empirical risk is mean squared error:

$$R_{\text{sq}}(h) = \frac{1}{n} \sum_{i=1}^n (y_i - h)^2$$

- When we use the absolute loss function, $L_{\text{abs}}(y_i, h) = |y_i - h|$, the corresponding empirical risk is mean absolute error:

$$R_{\text{abs}}(h) = \frac{1}{n} \sum_{i=1}^n |y_i - h|$$

Empirical risk minimization, in general

Key idea: If $L(y_i, h)$ is **any** loss function, the corresponding empirical risk is:

$$R(h) = \frac{1}{n} \sum_{i=1}^n L(y_i, h)$$

Question 🤔

Answer in chat or Q&A

Remember, you can always ask questions.

Question 🤔

Answer in chat (not Q&A)

$$R_{\text{sq}}(h) = \frac{1}{n} \sum_{i=1}^n (y_i - h)^2$$

$$R_{\text{abs}}(h) = \frac{1}{n} \sum_{i=1}^n |y_i - h|$$

Is the following statement true, for any dataset $y_1, y_2, \dots, y_n$ and prediction h ?

$$(R_{\text{abs}}(h))^2 = R_{\text{sq}}(h)$$

- A. It's true for any h and any dataset.
- B. It's can sometimes be true, but not in general.
- C. It's never true.

Choosing a loss function

Now what?

- We know that, for the constant model $H(x) = h$, the **mean** minimizes mean **squared** error.
- We also know that, for the constant model $H(x) = h$, the **median** minimizes mean **absolute** error.
- **How does our choice of loss function impact the resulting optimal prediction?**

Comparing the mean and median

- Consider our example dataset of 5 commute times.

$$y_1 = 72 \qquad y_2 = 90 \qquad y_3 = 61 \qquad y_4 = 85 \qquad y_5 = 92$$

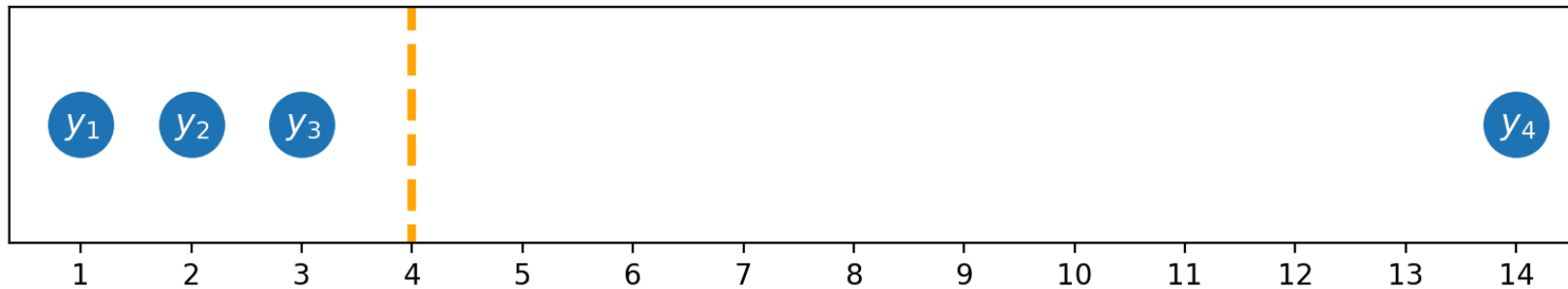
- As of now, the **median is 85** and the **mean is 80**.
- What if we add 200 to the largest commute time, 92?

$$y_1 = 72 \qquad y_2 = 90 \qquad y_3 = 61 \qquad y_4 = 85 \qquad y_5 = 292$$

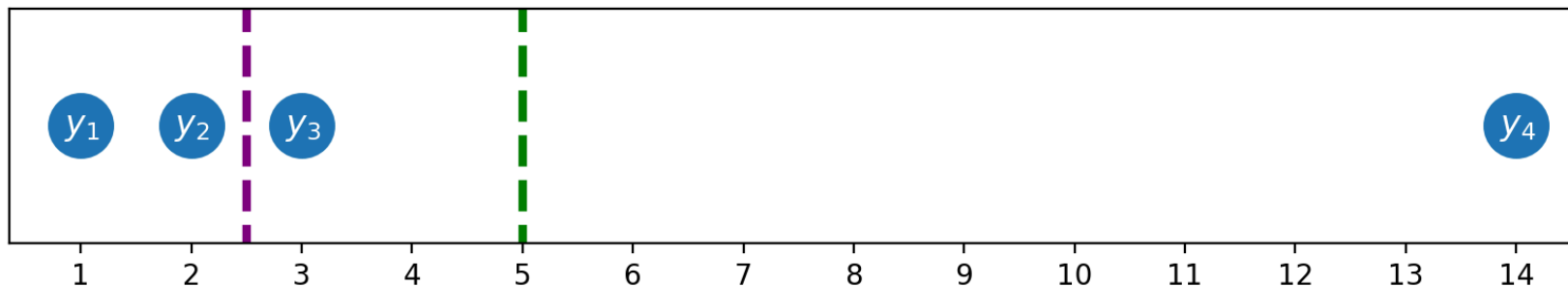
- Now, the median is 10 but the mean is 12.5!
- **Key idea:** The mean is quite **sensitive** to outliers.

Outliers

Below, $|y_4 - h|$ is 10 times as big as $|y_3 - h|$, but $(y_4 - h)^2$ is 100 times $(y_3 - h)^2$.

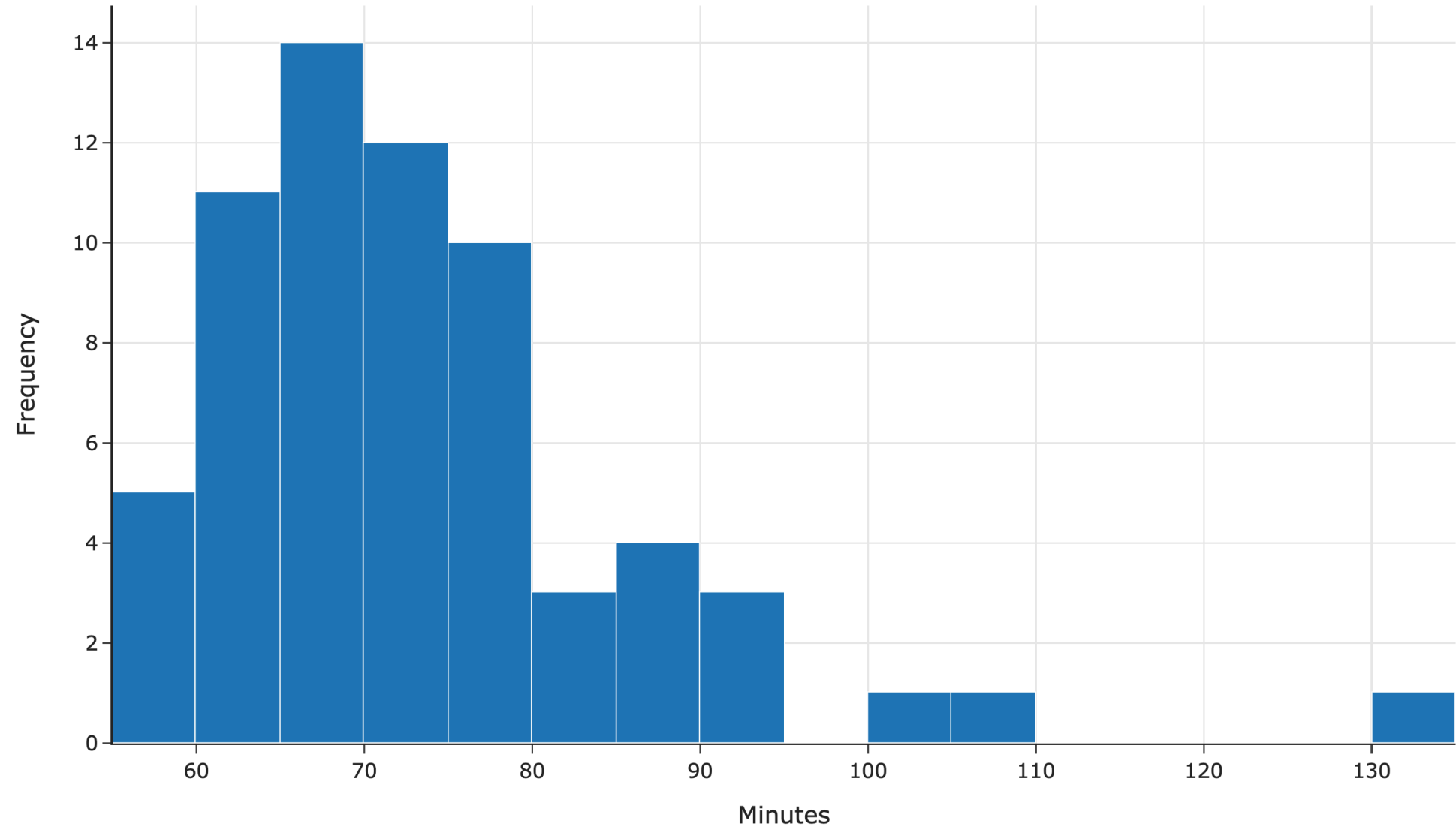


The result is that the **mean** is "pulled" in the direction of outliers, relative to the **median**.



As a result, we say the **median** is **robust** to outliers. But the **mean** was easier to solve for.

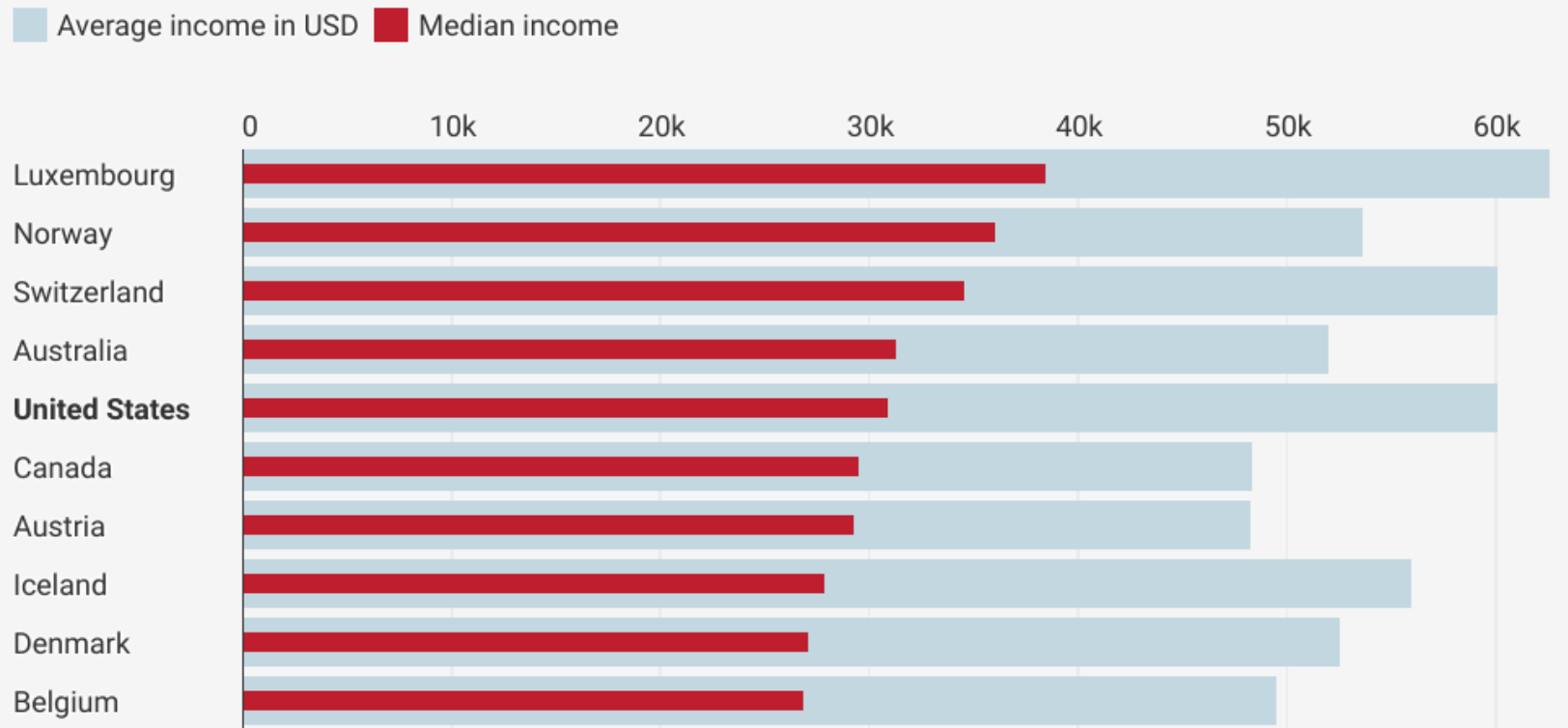
Distribution of Commuting Time



Example: Income inequality

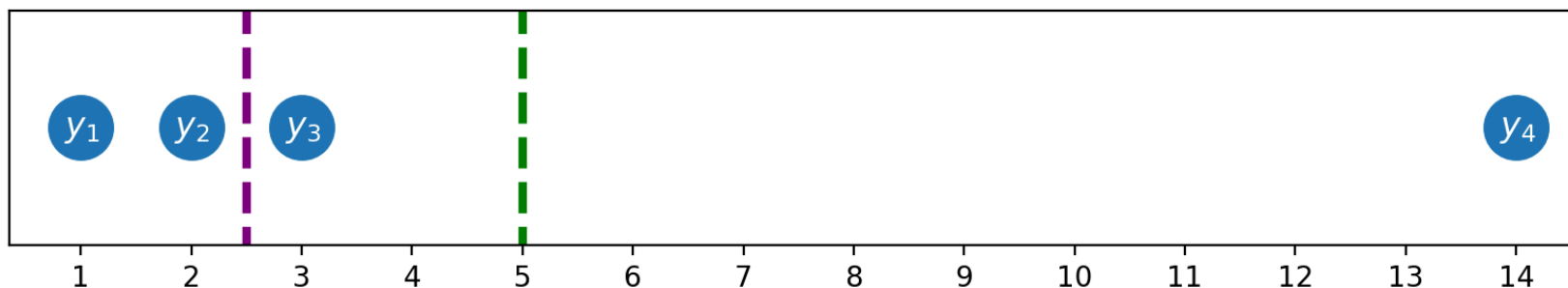
Average vs median income

Median and mean income between 2012 and 2014 in selected OECD countries, in USD; weighted by the currencies' respective [purchasing power](#) (PPP).



Balance points

Both the **mean** and **median** are "balance points" in the distribution.



- The **mean** is the point where $\sum_{i=1}^n (y_i - h) = 0$.
 - This appears in Homework 1!
- The **median** is the point where $\# (y_i < h) = \# (y_i > h)$.

Why stop at squared loss?

Empirical Risk, $R(h)$	Derivative of Empirical Risk, $\frac{d}{dh} R(h)$	Minimizer
$\frac{1}{n} \sum_{i=1}^n y_i - h $	$\frac{1}{n} \left(\sum_{y_i < h} 1 - \sum_{y_i > h} 1 \right)$	median
$\frac{1}{n} \sum_{i=1}^n (y_i - h)^2$	$\frac{-2}{n} \sum_{i=1}^n (y_i - h)$	mean
$\frac{1}{n} \sum_{i=1}^n y_i - h ^3$		???
$\frac{1}{n} \sum_{i=1}^n (y_i - h)^4$		???
$\frac{1}{n} \sum_{i=1}^n (y_i - h)^{100}$		???
...	...	...

Generalized L_p loss

For any $p \geq 1$, define the L_p loss as follows:

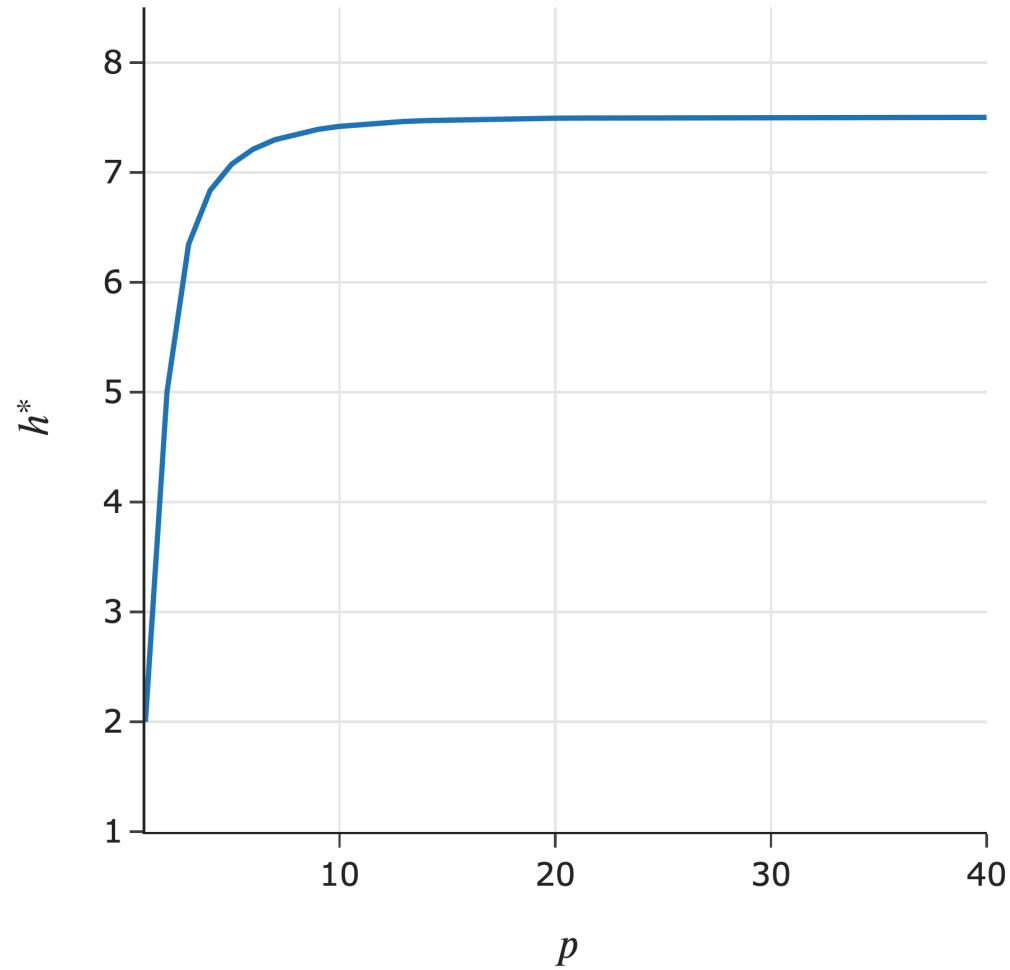
$$L_p(y_i, h) = |y_i - h|^p$$

The corresponding empirical risk is:

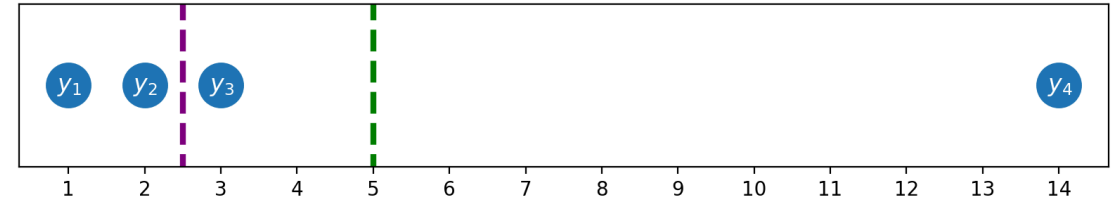
$$R_p(h) = \frac{1}{n} \sum_{i=1}^n |y_i - h|^p$$

- When $p = 1$, $h^* = \text{Median}(y_1, y_2, \dots, y_n)$.
- When $p = 2$, $h^* = \text{Mean}(y_1, y_2, \dots, y_n)$.
- What about when $p = 3$?
- What about when $p \rightarrow \infty$?

What value does h^* approach, as $p \rightarrow \infty$?



Consider the dataset 1, 2, 3, 14:



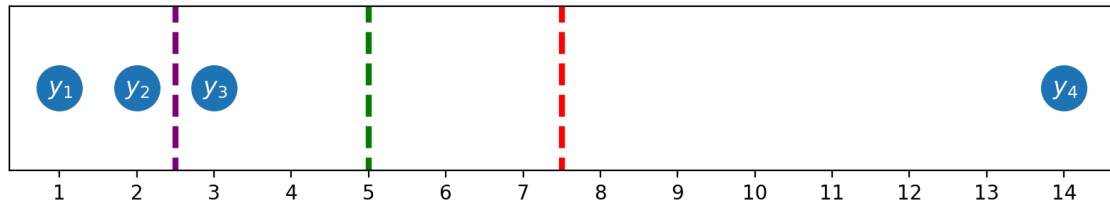
On the left:

- The x -axis is p .
- The y -axis is h^* , the optimal constant prediction for L_p loss:

$$h^* = \operatorname{argmin}_h \frac{1}{n} \sum_{i=1}^n |y_i - h|^p$$

The *midrange* minimizes average L_∞ loss!

On the previous slide, we saw that as $p \rightarrow \infty$, the minimizer of mean L_p loss approached **the midpoint of the minimum and maximum values in the dataset, or the midrange**.



- As $p \rightarrow \infty$, $R_p(h) = \frac{1}{n} \sum_{i=1}^n |y_i - h|^p$ minimizes the **"worst case" distance from any data point**". (Read more [here](#)).
- If your measure of "good" is "not far from any one data point", then the midrange is the best prediction.

Another example: 0-1 loss

Consider, for example, the **0-1 loss**:

$$L_{0,1}(y_i, h) = \begin{cases} 0 & y_i = h \\ 1 & y_i \neq h \end{cases}$$

The corresponding empirical risk is:

$$R_{0,1}(h) = \frac{1}{n} \sum_{i=1}^n L_{0,1}(y_i, h)$$

Question 🤔

Answer in chat (not Q&A)

$$R_{0,1}(h) = \frac{1}{n} \sum_{i=1}^n \begin{cases} 0 & y_i = h \\ 1 & y_i \neq h \end{cases}$$

Suppose $y_1, y_2, \dots, y_n$ are all unique. What is $R_{0,1}(y_1)$?

- A. 0.
- B. $\frac{1}{n}$.
- C. $\frac{n-1}{n}$.
- D. 1.

Minimizing empirical risk for 0-1 loss

$$R_{0,1}(h) = \frac{1}{n} \sum_{i=1}^n \begin{cases} 0 & y_i = h \\ 1 & y_i \neq h \end{cases}$$

Summary: Choosing a loss function

Key idea: Different loss functions lead to different best predictions, h^* !

Loss	Minimizer	Always Unique?	Robust to Outliers?	Differentiable?
L_{sq}	mean	yes 	no 	yes 
L_{abs}	median	no 	yes 	no 
L_{∞}	midrange	yes 	no 	no 
$L_{0,1}$	mode	no 	yes 	no 

The optimal predictions, h^* , are all **summary statistics** that measure the **center** of the dataset in different ways.

Center and spread

What does it mean?

- The general form of empirical risk, for any loss function $L(y_i, h)$, is:

$$R(h) = \frac{1}{n} \sum_{i=1}^n L(y_i, h)$$

- As we just saw, the input h^* that minimizes $R(h)$ is some measure of the **center** of the dataset.
 - Examples include the mean (L_{sq}), median (L_{abs}), and mode ($L_{0,1}$).
- The minimum output, $R(h^*)$, represents some measure of the **spread**, or variation, in the dataset.

Squared loss

- The empirical risk for squared loss, i.e. mean squared error, is:

$$R_{\text{sq}}(h) = \frac{1}{n} \sum_{i=1}^n (y_i - h)^2$$

- $R_{\text{sq}}(h)$ is minimized when $h^* = \text{Mean}(y_1, y_2, \dots, y_n)$.
- Therefore, the minimum value of $R_{\text{sq}}(h)$ is:

$$\begin{aligned} R_{\text{sq}}(h^*) &= R_{\text{sq}}(\text{Mean}(y_1, y_2, \dots, y_n)) \\ &= \frac{1}{n} \sum_{i=1}^n (y_i - \text{Mean}(y_1, y_2, \dots, y_n))^2 \end{aligned}$$

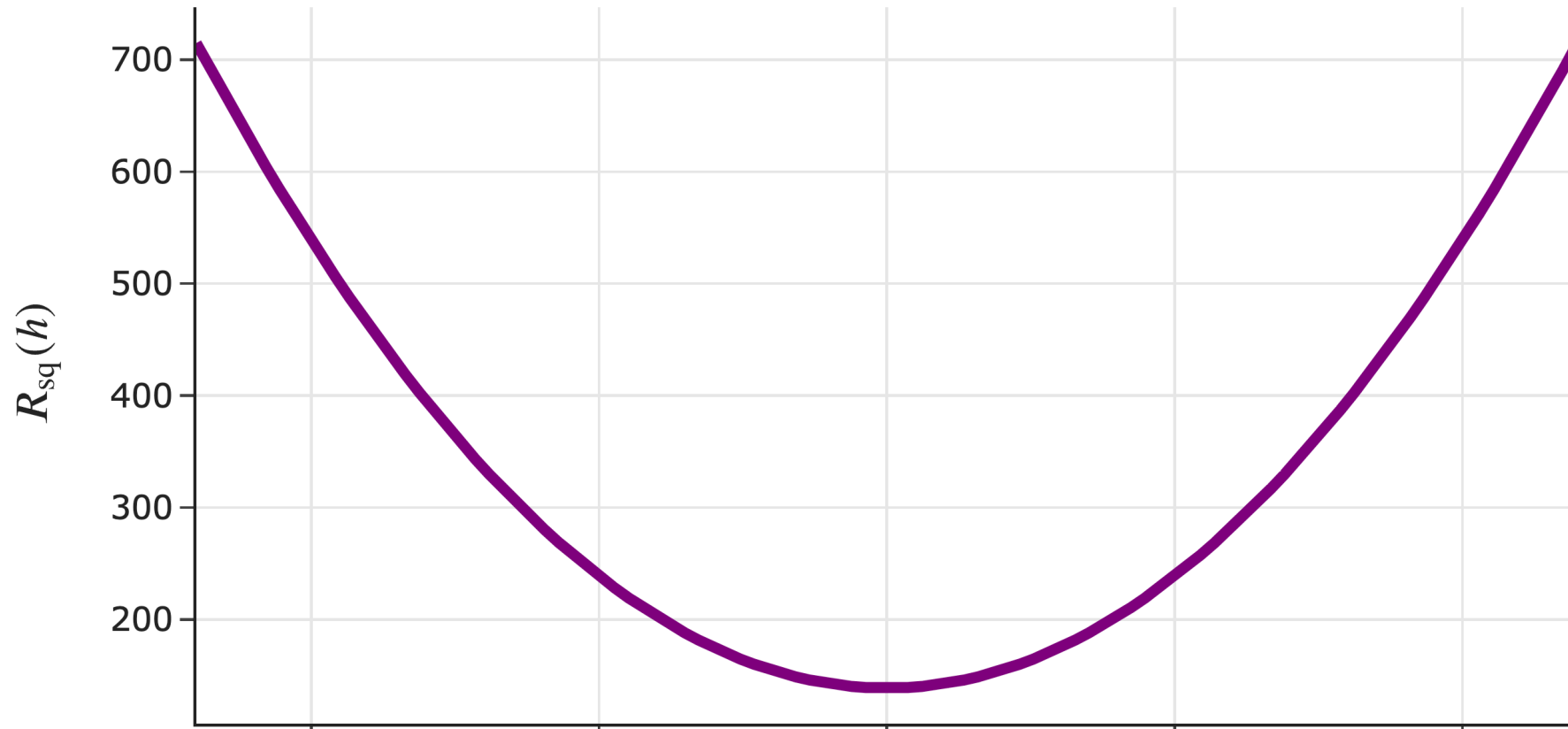
Variance

- The minimum value of $R_{sq}(h)$ is the mean squared deviation from the mean, more commonly known as the **variance**.

$$\text{Variance}(y_1, y_2, \dots, y_n) = \frac{1}{n} \sum_{i=1}^n (y_i - \text{Mean}(y_1, y_2, \dots, y_n))^2$$

- It measures the squared distance of each data point from the mean, on average.
- Its square root is called the **standard deviation**.

$$R_{\text{sq}}(h) = \frac{1}{5} \left((72 - h)^2 + (90 - h)^2 + (61 - h)^2 + (85 - h)^2 + (92 - h)^2 \right)$$



Absolute loss

- The empirical risk for absolute loss, i.e. mean absolute error, is:

$$R_{\text{abs}}(h) = \frac{1}{n} \sum_{i=1}^n |y_i - h|$$

- $R_{\text{abs}}(h)$ is minimized when $h^* = \text{Median}(y_1, y_2, \dots, y_n)$.
- Therefore, the minimum value of $R_{\text{abs}}(h)$ is:

$$\begin{aligned} R_{\text{abs}}(h^*) &= \frac{1}{n} \sum_{i=1}^n |y_i - h| \\ &= R_{\text{abs}}(h) = \frac{1}{n} \sum_{i=1}^n |y_i - \text{Median}(y_1, y_2, \dots, y_n)| \end{aligned}$$

Mean absolute deviation from the median

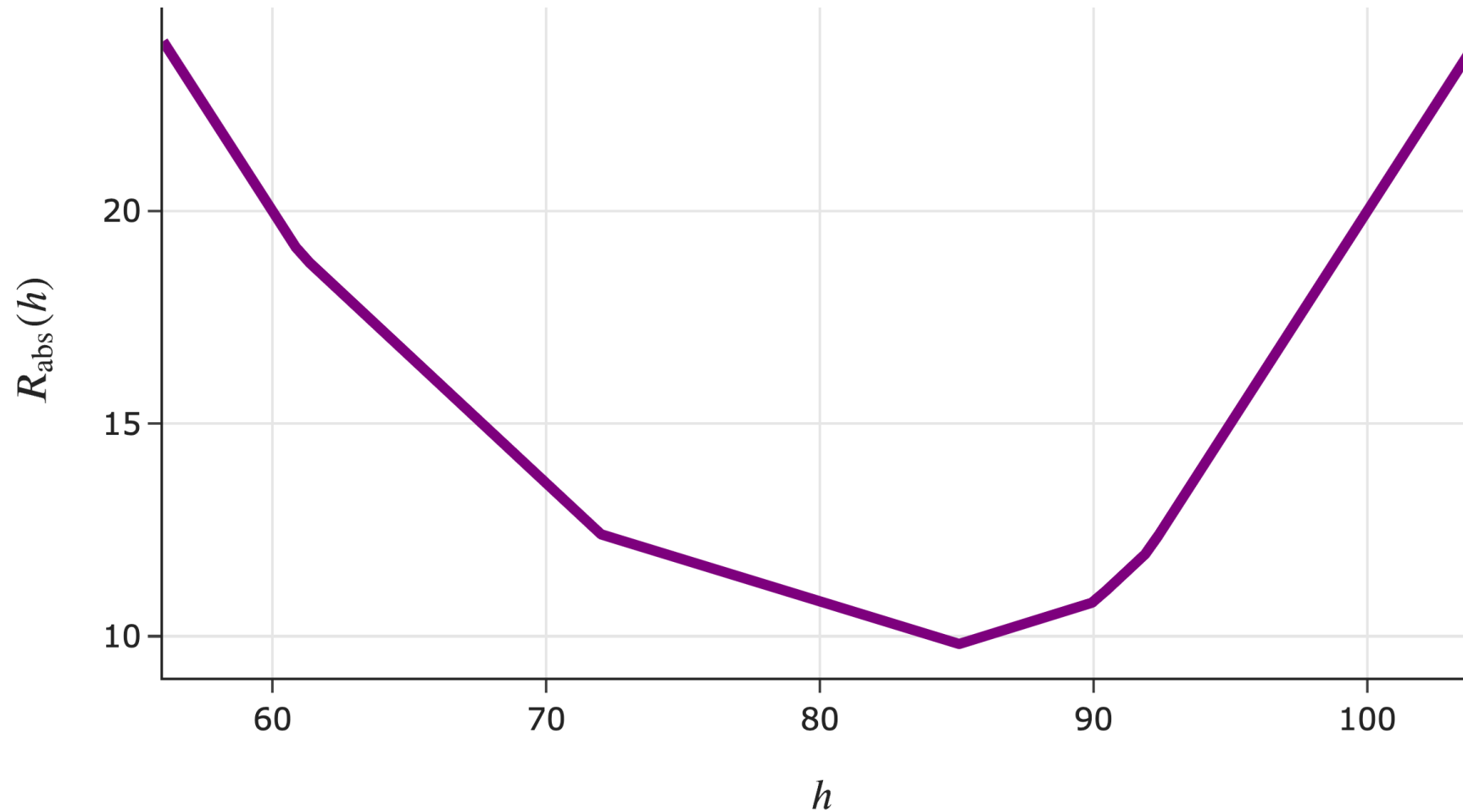
- The minimum value of $R_{\text{abs}}(h)$ is the **mean absolute deviation from the median**.

$$\text{MAD from the median}(y_1, y_2, \dots, y_n) = \frac{1}{n} \sum_{i=1}^n |y_i - \text{Median}(y_1, y_2, \dots, y_n)|$$

- It measures how far each data point is from the median, on average.
- **Example:** What's the MAD from the median in the dataset 2, 3, 3, 4, 5?

Mean absolute deviation from the median

$$R_{\text{abs}}(h) = \frac{1}{5}(|72 - h| + |90 - h| + |61 - h| + |85 - h| + |92 - h|)$$



0-1 loss

- The empirical risk for the 0-1 loss is:

$$R_{0,1}(h) = \frac{1}{n} \sum_{i=1}^n \begin{cases} 0 & y_i = h \\ 1 & y_i \neq h \end{cases}$$

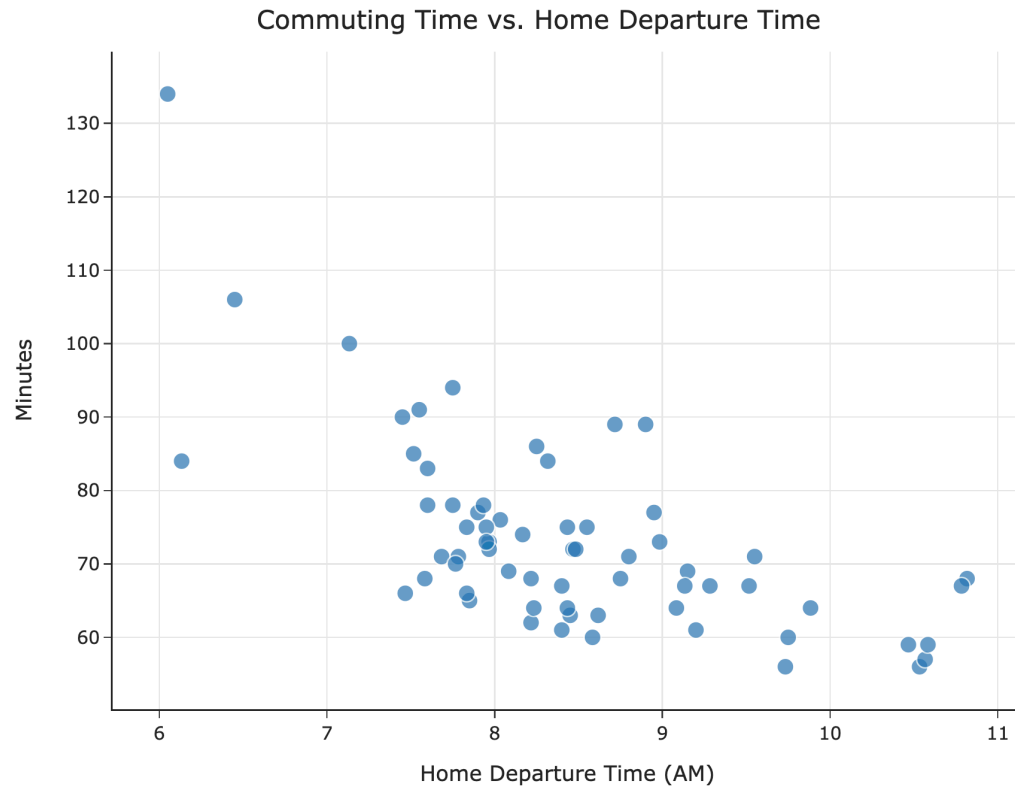
- This is the proportion (between 0 and 1) of data points not equal to h .
- $R_{0,1}(h)$ is minimized when $h^* = \text{Mode}(y_1, y_2, \dots, y_n)$.
- Therefore, $R_{0,1}(h^*)$ is the proportion of data points not equal to the mode.
- **Example:** What's the proportion of values not equal to the mode in the dataset 2, 3, 3, 4, 5?

Summary of center and spread

- Different loss functions $L(y_i, h)$ lead to different empirical risk functions $R(h)$, which are minimized at various measures of **center**.
- The minimum values of empirical risk, $R(h^*)$, are various measures of **spread**.
- There are many different ways to measure both center and spread; these are sometimes called **descriptive statistics**.

What's next?

Towards simple linear regression



- In Lecture 1, we introduced the idea of a hypothesis function, $H(x)$.
- We've focused on finding the best **constant model**, $H(x) = h$.
- Now that we understand the modeling recipe, we can apply it to find the best **simple linear regression model**, $H(x) = w_0 + w_1x$.
- This will allow us to make predictions that aren't all the same for every data point.

The modeling recipe

1. Choose a model.
2. Choose a loss function.
3. Minimize average loss to find optimal model parameters.

Break

Lecture 2 part 2

Simple Linear Regression

DSC 40A, Summer 2026

Part 2 Agenda

- Recap: Center and spread.
- Simple linear regression.
- Minimizing mean squared error for the simple linear model.

Question 🤔

Answer in chat; ask in Q&A

Remember, you can always ask questions!

Recap: Center and spread

The relationship between h^* and $R(h^*)$

- Recall, for a general loss function L and the constant model $H(x) = h$, empirical risk is of the form:

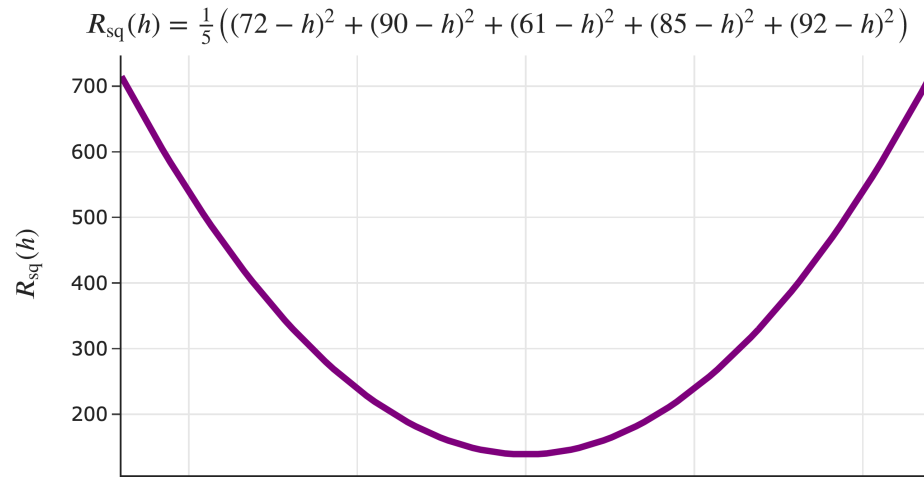
$$R(h) = \frac{1}{n} \sum_{i=1}^n L(y_i, h)$$

- h^* , the value of h that minimizes empirical risk, represents the **center** of the dataset in some way.
- $R(h^*)$, the smallest possible value of empirical risk, represents the **spread** of the dataset in some way.
- The specific center and spread depend on the choice of loss function.

Examples

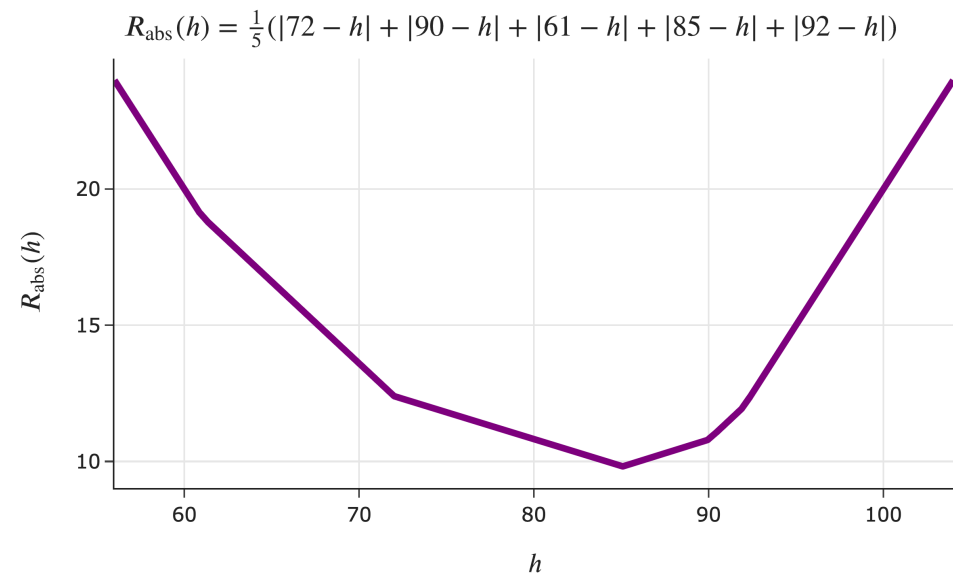
When using **squared loss**:

- $h^* = \text{Mean}(y_1, y_2, \dots, y_n)$.
- $R_{sq}(H^*) = \text{Variance}(y_1, y_2, \dots, y_n)$.



When using **absolute loss**:

- $h^* = \text{Median}(y_1, y_2, \dots, y_n)$.
- $R_{abs}(h^*) = \text{MAD from the median}$.



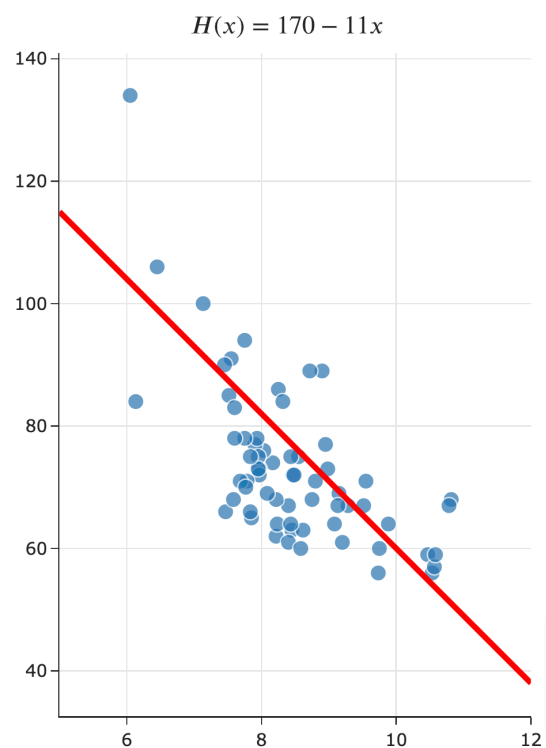
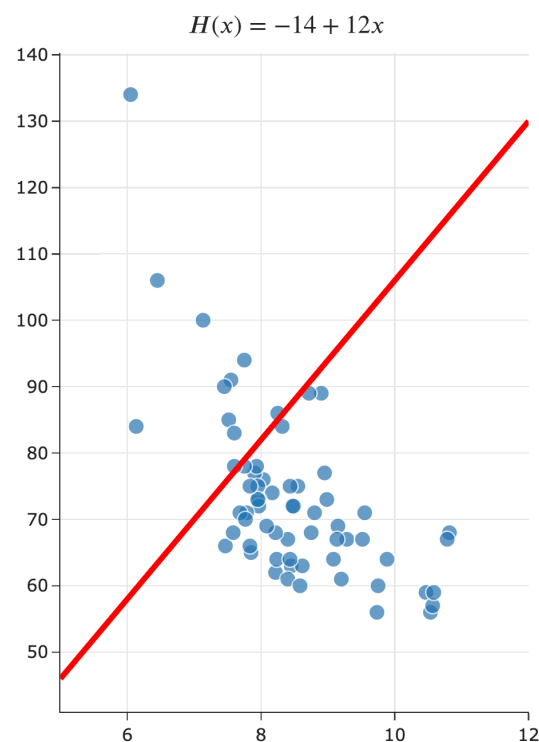
Simple linear regression

Recap: Hypothesis functions and parameters

A hypothesis function, H , takes in an x as input and returns a predicted y .

Parameters define the relationship between the input and output of a hypothesis function.

The simple linear regression model, $H(x) = w_0 + w_1x$, has two parameters: w_0 and w_1 .



The modeling recipe

1. Choose a model.
2. Choose a loss function.
3. Minimize average loss to find optimal model parameters.

Minimizing mean squared error for the simple linear model

- We'll choose squared loss, since it's the easiest to minimize.
- Our goal, then, is to find the linear hypothesis function $H^*(x)$ that minimizes empirical risk:

$$R_{\text{sq}}(H) = \frac{1}{n} \sum_{i=1}^n (y_i - H(x_i))^2$$

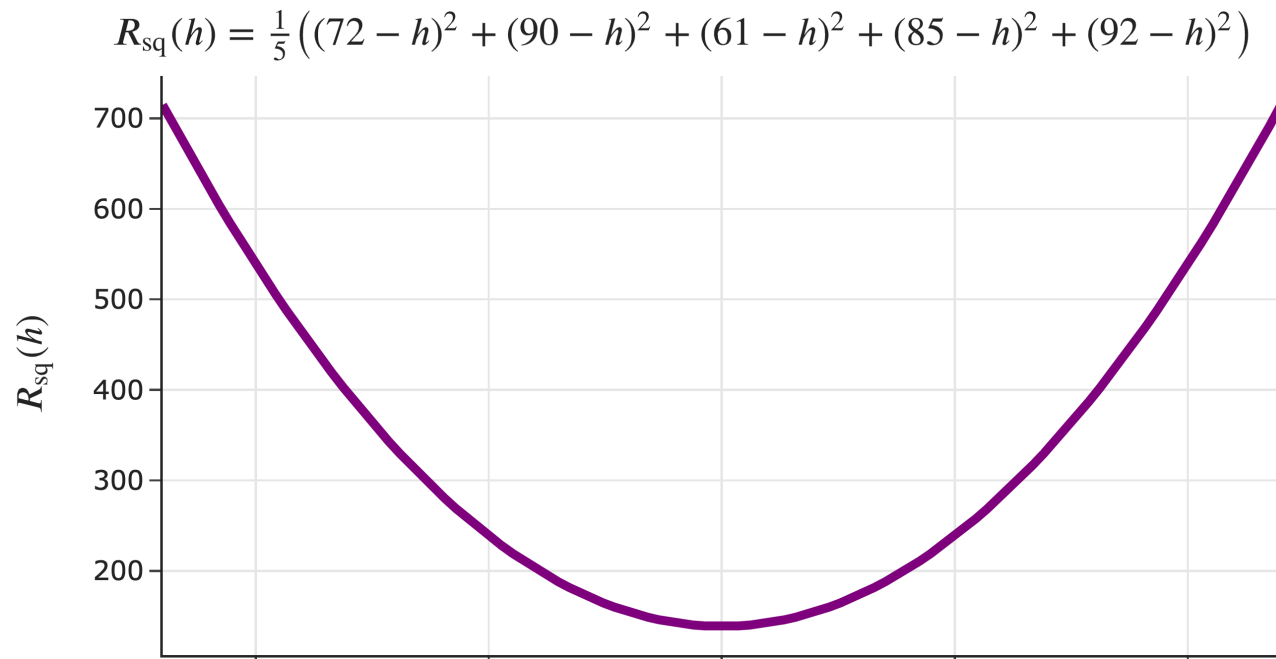
- Since linear hypothesis functions are of the form $H(x) = w_0 + w_1x$, we can re-write R_{sq} as a function of w_0 and w_1 :

$$R_{\text{sq}}(w_0, w_1) = \frac{1}{n} \sum_{i=1}^n (y_i - (w_0 + w_1x_i))^2$$

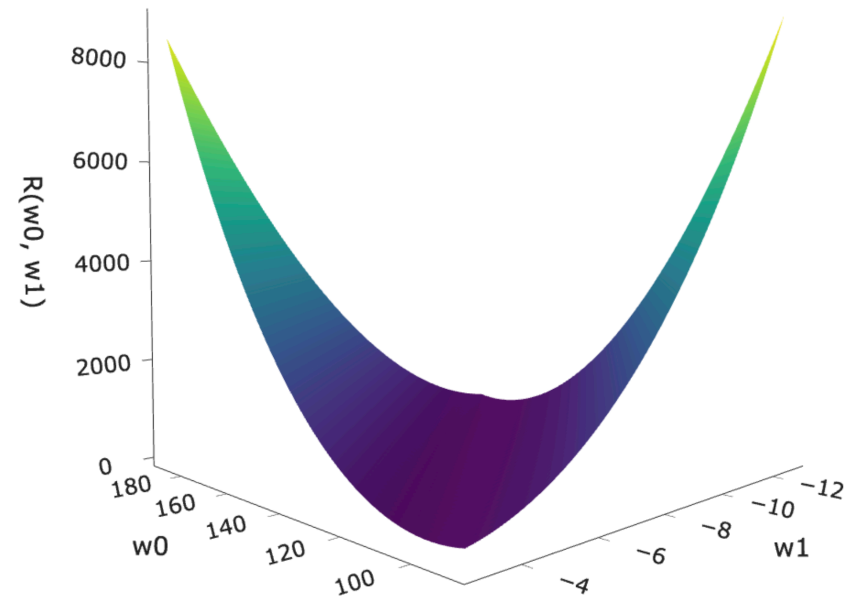
- How do we find the parameters w_0^* and w_1^* that minimize $R_{\text{sq}}(w_0, w_1)$?

Loss surface

For the constant model, the graph of $R_{\text{sq}}(h)$ looked like a parabola.



What does the graph of $R_{\text{sq}}(w_0, w_1)$ look like for the simple linear regression model?



Minimizing mean squared error for the simple linear model

Minimizing multivariate functions

- Our goal is to find the parameters w_0^* and w_1^* that minimize mean squared error:

$$R_{\text{sq}}(w_0, w_1) = \frac{1}{n} \sum_{i=1}^n (y_i - (w_0 + w_1 x_i))^2$$

- R_{sq} is a function of two variables: w_0 and w_1 .
- To minimize a function of multiple variables:
 - Take partial derivatives with respect to each variable.
 - Set all partial derivatives to 0.
 - Solve the resulting system of equations.
 - Ensure that you've found a minimum, rather than a maximum or saddle point (using the [second derivative test](#) for multivariate functions).

Example

Find the point (x, y, z) at which the following function is minimized.

$$f(x, y) = x^2 - 8x + y^2 + 6y - 7$$

Minimizing mean squared error

$$R_{\text{sq}}(w_0, w_1) = \frac{1}{n} \sum_{i=1}^n (y_i - (w_0 + w_1 x_i))^2$$

To find the w_0^* and w_1^* that minimize $R_{\text{sq}}(w_0, w_1)$, we'll:

1. Find $\frac{\partial R_{\text{sq}}}{\partial w_0}$ and set it equal to 0.
2. Find $\frac{\partial R_{\text{sq}}}{\partial w_1}$ and set it equal to 0.
3. Solve the resulting system of equations.

Question 🤔

Answer in chat

$$R_{\text{sq}}(w_0, w_1) = \frac{1}{n} \sum_{i=1}^n (y_i - (w_0 + w_1 x_i))^2$$

Which of the following is equal to $\frac{\partial R_{\text{sq}}}{\partial w_0}$?

- A. $\frac{1}{n} \sum_{i=1}^n (y_i - (w_0 + w_1 x_i))$
- B. $-\frac{1}{n} \sum_{i=1}^n (y_i - (w_0 + w_1 x_i))$
- C. $-\frac{2}{n} \sum_{i=1}^n (y_i - (w_0 + w_1 x_i)) x_i$
- D. $-\frac{2}{n} \sum_{i=1}^n (y_i - (w_0 + w_1 x_i))$

$$R_{\text{sq}}(w_0, w_1) = \frac{1}{n} \sum_{i=1}^n (y_i - (w_0 + w_1 x_i))^2$$

$$\frac{\partial R_{\text{sq}}}{\partial w_0} =$$

$$R_{\text{sq}}(w_0, w_1) = \frac{1}{n} \sum_{i=1}^n (y_i - (w_0 + w_1 x_i))^2$$

$$\frac{\partial R_{\text{sq}}}{\partial w_1} =$$

Strategy

We have a system of two equations and two unknowns (w_0 and w_1):

$$-\frac{2}{n} \sum_{i=1}^n (y_i - (w_0 + w_1 x_i)) = 0 \quad -\frac{2}{n} \sum_{i=1}^n (y_i - (w_0 + w_1 x_i)) x_i = 0$$

To proceed, we'll:

1. Solve for w_0 in the first equation.

The result becomes w_0^* , because it's the "best intercept."

2. Plug w_0^* into the second equation and solve for w_1 .

The result becomes w_1^* , because it's the "best slope."

Solving for w_0^*

$$-\frac{2}{n} \sum_{i=1}^n (y_i - (w_0 + w_1 x_i)) = 0$$

Solving for w_1^*

$$-\frac{2}{n} \sum_{i=1}^n (y_i - (w_0 + w_1 x_i)) x_i = 0$$

Least squares solutions

We've found that the values w_0^* and w_1^* that minimize R_{sq} are:

$$w_1^* = \frac{\sum_{i=1}^n (y_i - \bar{y})x_i}{\sum_{i=1}^n (x_i - \bar{x})x_i} \qquad w_0^* = \bar{y} - w_1^* \bar{x}$$

where:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \qquad \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$$

These formulas work, but let's re-write w_1^* to be a little more symmetric.

An equivalent formula for w_1^*

Claim:

$$w_1^* = \frac{\sum_{i=1}^n (y_i - \bar{y})x_i}{\sum_{i=1}^n (x_i - \bar{x})x_i} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Proof:

Least squares solutions

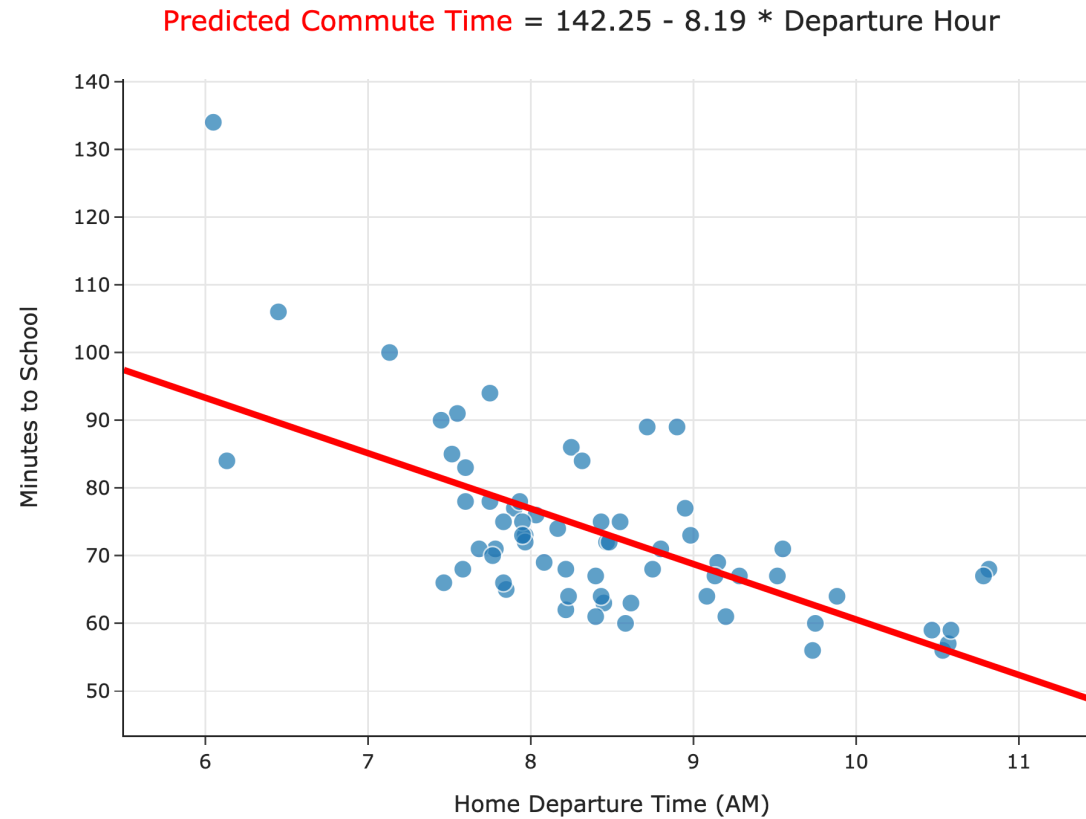
- The **least squares solutions** for the intercept w_0 and slope w_1 are:

$$w_1^* = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad w_0^* = \bar{y} - w_1^* \bar{x}$$

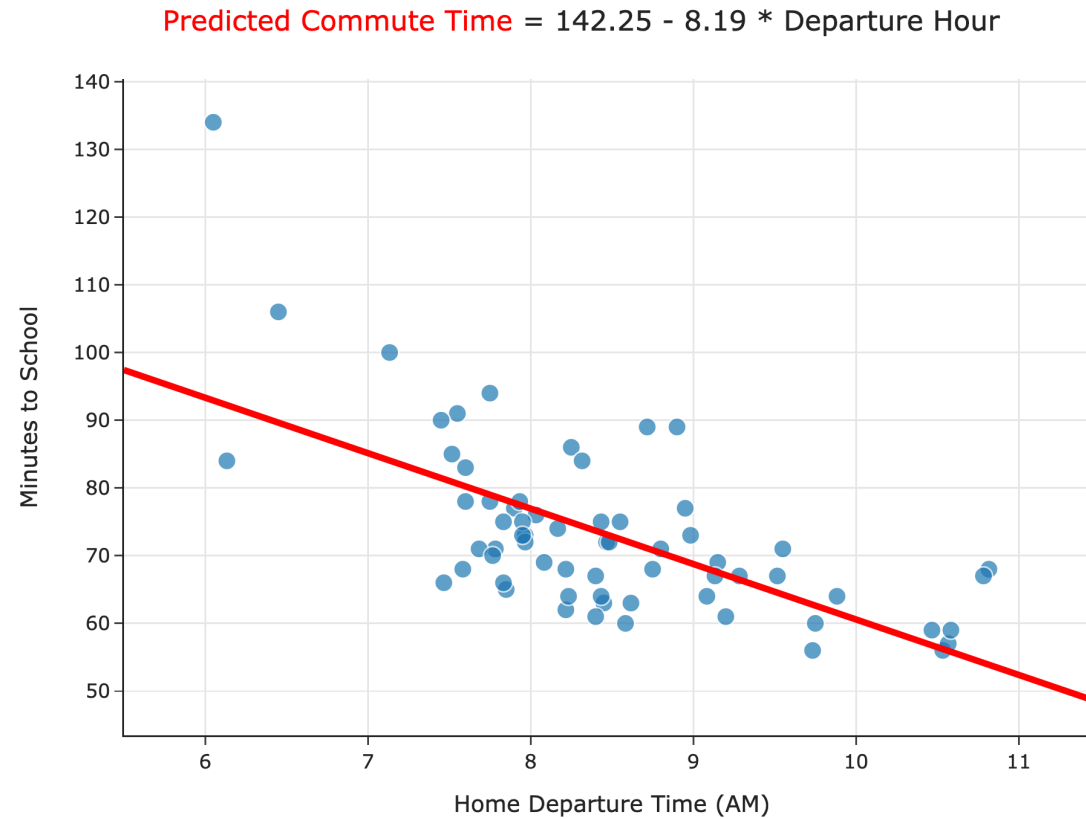
- We say w_0^* and w_1^* are **optimal parameters**, and the resulting line is called the **regression line**.
- The process of minimizing empirical risk to find optimal parameters is also called "**fitting to the data**."
- To make predictions about the future, we use $H^*(x) = w_0^* + w_1^*x$.

Let's test these formulas out in code! Follow along by downloading the files on Piazza.

[Google Colab](#) and [commute-times.csv](#)



Causality



Can we conclude that leaving later **causes** you to get to school quicker?

What's next?

We now know how to find the optimal slope and intercept for linear hypothesis functions.

Next, we'll:

- See how the formulas we just derived connect to the formulas for the slope and intercept of the regression line we saw in DSC 10.
 - They're the same, but we need to do a bit of work to prove that.
- Learn how to interpret the slope of the regression line.
- Discuss *causality*.
- Learn how to build regression models with **multiple inputs**.
 - To do this, we'll need linear algebra!